

Lateral Tool Transfer

Adversaries may transfer tools or other files between systems in a compromised environment. Once brought into the victim environment (i.e., [Ingress Tool Transfer](#)) files may then be copied from one system to another to stage adversary tools or other files over the course of an operation.

Adversaries may copy files between internal victim systems to support lateral movement using inherent file sharing protocols such as file sharing over [SMB/Windows Admin Shares](#) to connected network shares or with authenticated connections via [Remote Desktop Protocol](#).^[1]

Files can also be transferred using native or otherwise present tools on the victim system, such as scp, rsync, curl, sftp, and [ftp](#). In some cases, adversaries may be able to leverage [Web Services](#) such as Dropbox or OneDrive to copy files from one machine to another via shared, automatically synced folders.^[2]

ID: T1570

Sub-techniques: No sub-techniques

① **Tactic:** [Lateral Movement](#)

① **Platforms:** ESXi, Linux, Windows, macOS

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Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team moved their tools laterally within the corporate network and between the ICS and corporate network. ^[3]
C0025	2016 Ukraine Electric Power Attack	During the 2016 Ukraine Electric Power Attack , Sandworm Team used <code>move</code> to transfer files to a network share. ^[4]
C0034	2022 Ukraine Electric Power Attack	During the 2022 Ukraine Electric Power Attack , Sandworm Team used a Group Policy Object (GPO) to copy CaddyWiper 's executable <code>msserver.exe</code> from a staging server to a local hard drive before deployment. ^[5]
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks , the adversaries had placed the malicious payload on an accessible network share to facilitate propagation. ^{[6][7][8]}
G1030	Agrius	Agrius downloaded some payloads for follow-on execution from legitimate filesharing services such as <code>ufile.io</code> and <code>easyupload.io</code> . ^[9]
G1007	Aoqin Dragon	Aoqin Dragon has spread malware in target networks by copying modules to folders masquerading as removable devices. ^[10]
G0050	APT32	APT32 has deployed tools after moving laterally using administrative accounts. ^[11]
G0096	APT41	APT41 uses remote shares to move and remotely execute payloads during lateral movement. ^[12]
S0190	BITSAdmin	BITSAdmin can be used to create BITS Jobs to upload and/or download files from SMB file servers. ^[13]
G1043	BlackByte	BlackByte transferred tools such as Cobalt Strike and the AnyDesk remote access tool during operations using SMB shares. ^[14]
S1180	BlackByte Ransomware	BlackByte Ransomware spreads itself laterally by writing the JavaScript launcher file to mapped shared folders. ^[15]
S1068	BlackCat	BlackCat can replicate itself across connected servers via <code>psexec</code> . ^[16]
C0015	C0015	During C0015 , the threat actors used WMI to load Cobalt Strike onto additional hosts within a compromised network. ^[17]
C0018	C0018	During C0018 , the threat actors transferred the SoftPerfect Network Scanner and other tools to machines in the network using AnyDesk and PDQ Deploy. ^{[18][19]}
G0114	Chimera	Chimera has copied tools between compromised hosts using SMB. ^[20]
S0106	<code>cmd</code>	<code>cmd</code> can be used to copy files to/from a remotely connected internal system. ^[21]
S0062	DustySky	DustySky searches for network drives and removable media and duplicates itself onto them. ^[22]
G1003	Ember Bear	Ember Bear retrieves follow-on payloads direct from adversary-owned infrastructure for deployment on compromised hosts. ^[23]
S0367	Emotet	Emotet has copied itself to remote systems using the <code>service.exe</code> filename. ^[24]
S0404	esentutl	esentutl can be used to copy files to/from a remote share. ^[25]
S0361	Expand	Expand can be used to download or upload a file over a network share. ^[26]

ID	Name	Description
G0093	GALLIUM	GALLIUM has used PsExec to move laterally between hosts in the target network. ^[30]
S1229	Havoc	Havoc has the ability to copy files from one location to another. ^[31]
S0698	HermeticWizard	HermeticWizard can copy files to other machines on a compromised network. ^[32]
C0038	HomeLand Justice	During HomeLand Justice , threat actors initiated a process named Mellona.exe to spread the ROADSWEEP file encryptor and a persistence script to a list of internal machines. ^[33]
S0357	Impacket	Impacket has used its <code>wmiexec</code> command, leveraging Windows Management Instrumentation, to remotely stage and execute payloads in victim networks. ^[34]
G1032	INC Ransom	INC Ransom has used a rapid succession of copy commands to install a file encryption executable across multiple endpoints within compromised infrastructure. ^{[35][36]}
S1139	INC Ransomware	INC Ransomware can push its encryption executable to multiple endpoints within compromised infrastructure. ^[35]
S1132	IPsec Helper	IPsec Helper can download additional payloads from command and control nodes and execute them. ^[37]
S0372	LockerGoga	LockerGoga has been observed moving around the victim network via SMB, indicating the actors behind this ransomware are manually copying files form computer to computer instead of self-propagating. ^[1]
S0532	Lucifer	Lucifer can use certutil for propagation on Windows hosts within intranets. ^[38]
G0059	Magic Hound	Magic Hound has copied tools within a compromised network using RDP. ^[39]
G1051	Medusa Group	Medusa Group has utilized legitimate software services such as PDQ Deploy to transfer malicious binaries and tools to other victimized hosts within the target environment. ^[40]
S0457	Netwalker	Operators deploying Netwalker have used psexec to copy the Netwalker payload across accessible systems. ^[41]
S0365	Olympic Destroyer	Olympic Destroyer attempts to copy itself to remote machines on the network. ^[42]
C0014	Operation Wocao	During Operation Wocao , threat actors used SMB to copy files to and from target systems. ^[43]
S1017	OutSteel	OutSteel can download the Saint Bot malware for follow-on execution. ^[44]
S0029	PsExec	PsExec can be used to download or upload a file over a network share. ^[45]
S1242	Qilin	Qilin has used PsExec to distribute a second encryptor, named encryptor_1.exe, across the targeted environment. ^[46]
S9030	SameCoin	SameCoin can copy its wiper executable to remote machines within the same Active Directory. ^[47]
G0034	Sandworm Team	Sandworm Team has used <code>move</code> to transfer files to a network share and has copied payloads--such as Prestige ransomware--to an Active Directory Domain Controller and distributed via the Default Domain Group Policy Object. ^{[4][48]} Additionally, Sandworm Team has transferred an ISO file into the OT network to gain initial access. ^[5]
S0140	Shamoon	Shamoon attempts to copy itself to remote machines on the network. ^[49]
C0058	SharePoint ToolShell	During SharePoint ToolShell Exploitation , threat actors used Impacket to remotely stage and execute payloads in WMN. ^[50]

ID	Name	Description
G0010	Turla	Turla RPC backdoors can be used to transfer files to/from victim machines on the local network. ^{[53][54]}
G1048	UNC3886	UNC3886 has utilized Python scripts to transfer files between ESXi hosts and guest VMs. ^[55]
G1047	Velvet Ant	Velvet Ant transferred files laterally within victim networks through the Impacket toolkit. ^[34]
S1218	VIRTUALPIE	VIRTUALPIE has file transfer capabilities. ^[56]
S1217	VIRTUALPITA	VIRTUALPITA is capable of file transfer and arbitrary command execution. ^[56]
G1017	Volt Typhoon	Volt Typhoon has copied web shells between servers in targeted environments. ^[57]
S0366	WannaCry	WannaCry attempts to copy itself to remote computers after gaining access via an SMB exploit. ^[58]
G0102	Wizard Spider	Wizard Spider has used stolen credentials to copy tools into the <code>%TEMP%</code> directory of domain controllers. ^[59]

Mitigations

ID	Mitigation	Description
M1037	Filter Network Traffic	Consider using the host firewall to restrict file sharing communications such as SMB. ^[60]
M1031	Network Intrusion Prevention	Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary malware or unusual data transfer over known tools and protocols like FTP can be used to mitigate activity at the network level. Signatures are often for unique indicators within protocols and may be based on the specific obfuscation technique used by a particular adversary or tool, and will likely be different across various malware families and versions. ^[61]

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0183	Detection Strategy for Lateral Tool Transfer across OS platforms	AN0516	Correlate suspicious file transfers over SMB or Admin\$ shares with process creation events (e.g., cmd.exe, powershell.exe, certutil.exe) that do not align with normal administrative behavior. Detect remote file writes followed by execution of transferred binaries.
		AN0517	Monitor scp, rsync, curl, sftp, or ftp processes initiating transfers to internal systems combined with file creation events in unusual directories. Correlate transfer activity with subsequent execution of those binaries.
		AN0518	Detect anomalous use of scp, rsync, curl, or third-party sync apps transferring executables into user directories. Correlate new file creation with immediate execution events.
		AN0519	Identify lateral transfer via datastore file uploads or internal scp/ssh sessions that result in new VMX/VMDK or script files. Correlate transfer with VM execution or datastore modification.

References

1. Harbison, M. (2019, March 26). [Born This Way? Origins of LockerGoga](#). Retrieved April 16, 2019.
2. David Talbot. (2013, August 21). [Dropbox and Similar Services Can Sync Malware](#). Retrieved May 31, 2023.
3. Booz Allen Hamilton. (2016). [When The Lights Went Out](#). Retrieved December 18, 2024.
4. Joe Slowik. (2018, October 12). [Anatomy of an Attack: Detecting and Defeating CRASHOVERRIDE](#). Retrieved December 18, 2020.
5. Ken Proska, John Wolfram, Jared Wilson, Dan Black, Keith Lunden, Daniel Kapellmann Zafra, Nathan Brubaker, Tyler Mclellan, Chris Sistrunk. (2023, November 9). [Sandworm Disrupts Power in Ukraine Using a Novel Attack Against Operational Technology](#). Retrieved March 28, 2024.
6. CERT Polska. (2026, January 30). [Energy Sector Incident Report – 29 December](#). Retrieved April 22, 2026.
7. ESET. (2026, January 30). [Russian Sandworm group attacks energy company in Poland with DynoWiper, ESET Research discovers](#). Retrieved April 22, 2026.
8. ESET. (2026, January 30). [DynoWiper update: Technical analysis and attribution](#). Retrieved April 22, 2026.
9. Marc Salinas Fernandez & Jiri Vinopal. (2023, May 23). [AGRIUS DEPLOYS MONEYBIRD IN TARGETED ATTACKS AGAINST ISRAELI ORGANIZATIONS](#). Retrieved May 21, 2024.
10. Chen, Joey. (2022, June 9). [Aoqin Dragon I Newly-Discovered Chinese-linked APT Has Been Quietly Spying On Organizations For 10 Years](#). Retrieved July 14, 2022.
11. Dahan, A. (2017). [Operation Cobalt Kitty](#). Retrieved December 27, 2018.
12. Nikita Rostovcev. (2022, August 18). [APT41 World Tour 2021 on a tight schedule](#). Retrieved February 22, 2024.
13. Microsoft. (2019, July 12). [About BITS](#). Retrieved March 16, 2020.
14. Huseyin Can Yuceel. (2022, February 21). [TTPs used by BlackByte Ransomware Targeting Critical Infrastructure](#). Retrieved December 16, 2024.
15. Rodel Mendrez & Lloyd Macrohon. (2021, October 15). [BlackByte Ransomware – Pt. 1 In-depth Analysis](#). Retrieved December 16, 2024.
16. Microsoft Defender Threat Intelligence. (2022, June 13). [The many lives of BlackCat ransomware](#). Retrieved December 20, 2022.
17. DFIR Report. (2021, November 29). [CONTInuing the Bazar Ransomware Story](#). Retrieved September 29, 2022.
18. Venere, G. Neal, C. (2022, June 21). [Avos ransomware group expands with new attack arsenal](#). Retrieved January 11, 2023.
19. Costa, F. (2022, May 1). [RaaS AvosLocker Incident Response Analysis](#). Retrieved January 11, 2023.
20. Jansen, W. (2021, January 12). [Abusing cloud services to fly under the radar](#). Retrieved September 12, 2024.
21. Microsoft. (n.d.). [Copy](#). Retrieved April 26, 2016.
22. ClearSky. (2016, January 7). [Operation DustySky](#). Retrieved January 8, 2016.
23. Microsoft Threat Intelligence. (2023, June 14). [Cadet Blizzard emerges as a novel and distinct Russian threat actor](#). Retrieved July 10, 2023.
24. Binary Defense. (n.d.). [Emotet Evolves With new Wi-Fi Spreader](#). Retrieved September 8, 2023.
32. ESET. (2022, March 1). [IsaacWiper and HermeticWizard: New wiper and worm targeting Ukraine](#). Retrieved April 10, 2022.
33. CISA. (2022, September 23). [AA22-264A Iranian State Actors Conduct Cyber Operations Against the Government of Albania](#). Retrieved August 6, 2024.
34. Sygnia Team. (2024, June 3). [China-Nexus Threat Group ‘Velvet Ant’ Abuses F5 Load Balancers for Persistence](#). Retrieved March 14, 2025.
35. Team Huntress. (2023, August 11). [Investigating New INC Ransom Group Activity](#). Retrieved June 5, 2024.
36. Counter Threat Unit Research Team. (2024, April 15). [GOLD IONIC DEPLOYS INC RANSOMWARE](#). Retrieved June 5, 2024.
37. Amitai Ben & Shushan Ehrlich. (2021, May). [From Wiper to Ransomware: The Evolution of Agrius](#). Retrieved May 21, 2024.
38. Hsu, K. et al. (2020, June 24). [Lucifer: New Cryptojacking and DDoS Hybrid Malware Exploiting High and Critical Vulnerabilities to Infect Windows Devices](#). Retrieved November 16, 2020.
39. DFIR Report. (2021, November 15). [Exchange Exploit Leads to Domain Wide Ransomware](#). Retrieved January 5, 2023.
40. Threat Hunter Team Symantec and Carbon Black. (2025, March 6). [Medusa Ransomware Activity Continues to Increase](#). Retrieved October 15, 2025.
41. Szappanos, G., Brandt, A.. (2020, May 27). [Netwalker ransomware tools give insight into threat actor](#). Retrieved May 27, 2020.
42. Mercer, W. and Rascagneres, P. (2018, February 12). [Olympic Destroyer Takes Aim At Winter Olympics](#). Retrieved March 14, 2019.
43. Dantzig, M. v., Schamper, E. (2019, December 19). [Operation Wocao: Shining a light on one of China’s hidden hacking groups](#). Retrieved October 8, 2020.
44. Unit 42. (2022, February 25). [Spear Phishing Attacks Target Organizations in Ukraine, Payloads Include the Document Stealer OutSteel and the Downloader SaintBot](#). Retrieved June 9, 2022.
45. Russinovich, M. (2004, June 28). [PsExec](#). Retrieved December 17, 2015.
46. Takeda, T. et al. (2025, October 26). [Uncovering Qilin attack methods exposed through multiple cases](#). Retrieved March 26, 2026.
47. Check Point. (2024, November 12). [Hamis-affiliated Threat Actor WIRTE Continues its Middle East Operations and Moves to Disruptive Activity](#). Retrieved April 20, 2026.
48. MSTIC. (2022, October 14). [New “Prestige” ransomware impacts organizations in Ukraine and Poland](#). Retrieved January 19, 2023.
49. Falcone, R.. (2016, November 30). [Shamoon 2: Return of the Distrack Wiper](#). Retrieved January 11, 2017.
50. Microsoft Threat Intelligence. (2025, July 22). [Disrupting active exploitation of on-premises SharePoint vulnerabilities](#). Retrieved October 15, 2025.
51. Tyler McGraw, Thomas Elkins, and Evan McCann. (2024, May 10). [Ongoing Social Engineering Campaign Linked to Black Basta Ransomware Operators](#). Retrieved January 31, 2025.
52. Nicolas Falliere, Liam O Murchu, Eric Chien 2011, February. [W32.Stuxnet Dossier \(Version 1.4\)](#) Retrieved November 17, 2024.
53. Faou, M. and Dumont R.. (2019, May 29). [A dive into Turla PowerShell usage](#). Retrieved June 14, 2019.

31. [Ungur, P. \(n.d.\). HAVOC. Retrieved August 4, 2025.](#)
56. [Alexander Marvi, Jeremy Koppen, Tufail Ahmed, and Jonathan Lepore. \(2022, September 29\). Bad VIB\(E\)s Part One: Investigating Novel Malware Persistence Within ESXi Hypervisors. Retrieved March 26, 2025.](#)
57. [Counter Threat Unit Research Team. \(2023, May 24\). Chinese Cyberespionage Group BRONZE SILHOUETTE Targets U.S. Government and Defense Organizations. Retrieved July 27, 2023.](#)
58. [Noerenberg, E., Costis, A., and Quist, N. \(2017, May 16\). A Technical Analysis of WannaCry Ransomware. Retrieved December 8, 2024.](#)
59. [John, E. and Carvey, H. \(2019, May 30\). Unraveling the Spiderweb: Timelining ATT&CK Artifacts Used by GRIM SPIDER. Retrieved May 12, 2020.](#)
60. [Microsoft. \(2020, March 10\). Preventing SMB traffic from lateral connections and entering or leaving the network. Retrieved June 1, 2020.](#)
61. [Gardiner, J., Cova, M., Nagaraja, S. \(2014, February\). Command & Control Understanding, Denying and Detecting. Retrieved April 20, 2016.](#)